VICTORIA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2025
H2 GEOGRAPHY PAPER 1

MARK SCHEME

Qn	Answer	Marks
1a	**Illustrate interconnectedness in the global economy.** Award 1 mark for definition of global economy. Award 1 mark for illustration of interconnectedness in the global economy and a maximum of 1 additional mark for further development. Define: The global economy is a set of human activities and institutions linked together in the production, distribution, exchange and consumption of goods and services. [1] Students may choose to illustrate interconnectedness in the global economy in different ways: **Interconnectedness in financial markets** [1]: Countries implemented lockdowns to control the spread of COVID-19, which led to the shutdown of factories, transport routes and labour shortages, which eventually led to a global economic standstill. [1] **Interconnectedness in global supply chains** [1]: Manufacturing and assembly are highly interdependent, relying on global suppliers, specialized labour, and precise scheduling. A disruption at any point in the chain can ripple across countries and industries, affecting the global supply of goods. When supplies of car components are in shortage, this leads to eventual delays in assembly and delivery of the finished vehicle. [1] **Interconnectedness in Global Production Networks** [1]: TNCs have an extensive GPN, and produce their goods in multiple countries to take advantage of cost advantages. In the manufacturing of a car for instance, raw materials, component parts and assembly take place in different countries. [1] **AO1**	3
1b	**Describe the global distribution of rare earth reserves and volume of rare earth metals extracted in 2020 in Resource 1.** Award 1 mark for each substantiated observation. • China leads both in production and reserves holding over a third (38%) of the world's rare earth reserves and extracting more than half the world's supply. [1] • Other significant players in reserves are Vietnam and Brazil at 19% and 18% respectively [1] though they are very minor in extraction [1].	5

	• There are other countries with rare earth reserves but do not have any extraction e.g. Greenland and Canada. [1] • There are also countries with extraction but unknown reserves of raw earth e.g. Thailand and Burundi. [1] **AO2**	
1c	**Explain the gap between reserves available and volume of rare earth metals extracted in Resource 1.** Award 1 mark for reason and up to a maximum of 2 marks for further development of reason. **Economic factors** Mining may not be profitable if global prices are low, demand is uncertain, or extraction costs are high. Example: A country like Brazil may have large reserves, but if rare earth prices drop or production costs (labour, infrastructure) are high, companies may delay mining projects. **Government intervention** Governments may restrict mining to protect ecosystems, comply with environmental regulations, or avoid geopolitical risks. Strategic stockpiling- Some countries deliberately limit production to conserve resources for future use or to maintain strategic leverage in global supply chains. **Infrastructure and Capacity** Technical limitations — such as lack of processing facilities, skilled workforce, or transportation infrastructure — can also limit production even when reserves are large. **AO1**	6
1d	**Describe the trends in global demand for rare earth metals in Resource 2.** Award 1 mark for each substantiated observation. **Overall growth:** • 2023 → 2050: grows from ~90 kt to ~200 kt (~2.2X increase). **Wind energy** • 2023 → 2050: grows from ~10 kt to ~25 kt (~2.5X increase). • Steady growth, but smaller compared to EVs. **Electric vehicles (EVs)** • 2023 → 2050: grows from ~5 kt to ~65 kt (~13X increase).	4

	• Strongest relative growth of all categories — EVs become the main growth driver for rare earths. **Others** • In 2023, "other uses" dominate total demand. By 2050, the share from EVs and wind energy is much larger, indicating a strong shift toward clean energy technologies as the main consumers of rare earths **AO2**	
1e	**With reference to Resource 3, explain three ways in which governments may influence the global car industry.** Award 1 mark for each role identified and up to a maximum of 2 marks for further development of reason. Candidate may suggest either 2 or 3 roles the government may play. • **Trade Agreements:** By negotiating favorable trade agreements, governments can help ensure stable access to critical minerals and mitigate supply chain disruptions. • **Investment in infrastructure:** Governments can provide financial support for the development of domestic mining and processing facilities for REEs. This investment can in the long run reduce reliance on imports and strength local supply chains. • **Strategic Partnerships:** Governments can engage in international collaborations to secure diversified supply chains for REEs. This includes partnerships with countries rich in rare earth resources. • **Crisis mitigation:** Governments can develop contingency plans to address potential supply disruptions, ensuring that the auto industry can adapt quickly to changes in the availability of critical resources • **Diplomatic Engagement:** Active diplomatic efforts can help resolve trade disputes and ensure that supply chains remain open and efficient. **AO2**	6
1f	**Explain how a multilateral organisation like the World Trade Organisation (WTO) may intervene in Resource 3.** Award 1 mark for each intervention identified and up to a maximum of 2 marks for further development of intervention. Candidate may suggest either 2 or 3 roles the WTO may play. **Facilitating Trade Negotiations** • The WTO can provide a platform for member countries to engage in negotiations surrounding trade agreements that address concerns about	6

	the export restrictions imposed by China. By fostering dialogue, the WTO can help reduce tensions and promote more stable trade conditions for critical minerals. **Dispute Resolution** • If a country believes that another member is violating trade agreements—such as those related to export controls or market access—the WTO can facilitate a dispute resolution process. This is crucial in cases where countries accuse others of unfair trade practices, as seen in the U.S. accusations against China. **Capacity Building** • The WTO can assist developing countries in building their capacity to engage in international trade effectively. This includes providing technical assistance and resources to help these countries establish their own rare earth production and processing facilities, thus diversifying the global supply chain. **AO1**	
2a	**Describe the characteristics of the Mekong Delta as seen in Resource 4.** Award 1 mark for each point and a maximum of 1 additional mark for further development of each point. Any 4 characteristics: - The delta has a fan-shape/arcuate shape. [1] - There is a wide delta plain approximately 250km in width. [1] - The delta plain is mostly covered in dense vegetation. [1] - There are multiple distributaries on the delta plain. [1] - The distributaries flow in a southeast direction towards the sea. [1] - The delta front is visible off the coast of the delta plain in shallow water where there is evidence of high sediment load being deposited. [1] **AO2**	4
2b	**Explain why the shape and size of deltas vary across space**. Award 1 mark for each point and a maximum of 1 additional mark for further development of each point. Students should discuss at least two out of the three types of deltas in detail. - A delta's shape and size are dependent on the interaction between the geomorphic processes of river, tide and wave action at the coastline where the river meets the sea. [1] - River-dominated deltas form where there is minimal wave action [1], thus allowing the deposited sediments to protrude further out into sea in an elongate shape. [1] - Wave- dominated deltas form where high wave action prevents the delta from protruding [1] but instead causes it to grow in the direction of the longshore drift parallel to the shoreline in a cuspate or arcuate shape. [1]	5

	- Tide-dominated deltas form where strong tidal currents [1] moving in and out of the river mouth arrange sediments into elongated bars perpendicular to the shoreline. [1] **AO1**	
2c	**With reference to Resource 5, explain how El Nino would affect rainfall patterns of the Mekong Delta.** Award 1 mark for each point and a maximum of 1 additional mark for further development of each point. - Under El Nino conditions, easterly trade winds weaken or reverse directions. [1] - This causes warm water to spread to the eastern Pacific. [1] - In the western Pacific, which is where the Mekong Delta is, cooler water will be left behind. [1] - The cooler water causes the air above it to cool and sink, forming a high-pressure area. [1] - The descending air inhibits rain formation, creating dry conditions for the Mekong Delta with much less rainfall as compared to normal. [1] **AO2**	5
2d	**Explain three conditions required for tropical cyclones to form**. Award 1 mark for each point and a maximum of 1 additional mark for further development of each point. - Tropical cyclones require <u>warm ocean waters</u>. [1] Sea surface temperatures should be sufficient high (26-27°C) to warm the air above it, causing it to rise to form a low-pressure zone. [1] This usually happens in the late summer months in each hemisphere. [1] - Tropical cyclones require a <u>sufficiently strong Coriolis effect</u>. [1] The deflection of winds as they are drawn into the low-pressure centre allows a vortex to be formed and maintained. [1] The Coriolis effect also causes the storm to spin and give rise to its rotating motion. [1] - Tropical cyclones require <u>weak wind shear</u>. [1] When wind shear is too high, the winds will disrupt the development of vertical cumulonimbus clouds that form the rain bands of a tropical cyclone. [1] The strong wind shear will also disperse heat from the warm core. [1] **AO1**	6
2e	**Describe the distribution of mangrove forests on the Mekong Delta as seen in Resource 6.** Award 1 mark for each point and a maximum of 1 additional mark for further development of each point. Any 4 points:	4

	- Mangrove forests line the entire coastline of the Mekong Delta. [1] - The largest mangrove forests are found in the southwest and the northeast of the delta, in Cape Ca Mau and Can Gio respectively. [1] - The mangrove forest in Cape Ca Mau is almost 50km wide, while the mangrove forest in Can Gio is approximately 25km wide. [1] - No mangrove forests are found upstream of the distributaries. [1] **AO2**	
2f	**With reference to Resource 7, describe the impacts of tropical cyclones on the ecosystem services of the Mekong Delta.** Award 1 mark for each point and a maximum of 1 additional mark for further development of each point. The storm surges and strong winds of tropical cyclones would damage the mangrove forests [1] which would negatively impact: - [Regulating ecosystem services] With fewer mangrove plants available to bind the soil together and stabilise the coastline [1], the delta would be less capable of protecting the land against waves and sea level rise. [1] - [Regulating ecosystem services] With fewer plants, less atmospheric CO2 will be captured and stored [1], thus undermining its function as a carbon sink and as a green lung of HCM city. [1] - [Supporting/Cultural] Damage to the mangrove forests would threaten the habitats of the king cobra, saltwater crocodile and spot-billed pelican. [1] The drop in biodiversity could in turn reduce its ability to function as an important educational or research site. [1] **AO2**	6
3	**The impacts of extractive industries in countries in low levels of development are always negative.' Evaluate this statement.** Candidates could approach the question by making a judgement on whether the impacts of extractive industries in LDCs are always negative in LDCs through a consideration of both positive and negative impacts (social, economic, environmental). Candidates could compare a range of extractive industries that may differ in technology used, scale and location, which would affect the extent of negative impacts. Candidates could also approach the question by making a judgment on whether the impacts are always negative through consideration of two or more case studies. Candidates could evaluate the contextual factors that could allow the negative impacts to be mitigated or minimized (e.g. community engagement, strong monitoring and enforcement of environmental regulations). Levels marked using Generic Level Descriptors for H2 essays **AO3**	20

4	**The development of the global economy has led to increasing variations in levels of development across space." Evaluate this statement.** Candidates could approach the question by making a judgement on whether the development of the global economy has increased variations in levels of development on a national or global scale. Candidates could consider how uneven flows of trade, capital and labor flows (e.g. NIDL) have contributed to increasing variations at different scales. Candidates could also use two or more case studies to illustrate how different actors in the global economy (e.g. states, TNCs, multilateral institutions) may influence these flows in a way that may increase or decrease such variations. Levels marked using Generic Level Descriptors for H2 essays **AO3**	20
5	**To what extent can fluvial floods be effectively managed? Evaluate this statement.** Candidates could approach the question by making a judgement on whether fluvial floods can be effectively managed by humans through a consideration of the natural and human factors contributing to the occurrence of floods. Candidates could also evaluate which factor is most significant in contributing to the occurrence of floods and if they are largely beyond the control of humans. Candidates could also approach the question by making a judgment on whether fluvial floods can be controlled by humans through a consideration of two or more case studies. Candidates could evaluate whether the severity of the impacts of each case study could be mitigated by humans. Candidates could also consider the contextual factors such as the nature of the flood and preparedness of the population, which may have contributed to differences in the extent the floods could be controlled. Levels marked using Generic Level Descriptors for H2 essays **AO3**	20
6	**Evaluate the factors influencing the formation of karst landforms in the humid tropics.** Candidates could approach the question by making a judgement on which factor is the most significant in influencing the formation of karst landforms. They could weigh the relative influence of natural (climate, geology, vegetation) and human factors on the development of karst landscapes. Candidates could also approach the question by making a judgment on whether some factors have a greater influence in some contexts than others through a consideration of two or more case studies. Candidates could evaluate the relative significance of the different factors in each case study and analyse the interplay between the factors.	20

Levels marked using Generic Level Descriptors for H2 essays

Generic Level Descriptors for H2 Essays		
Level	**Marks**	**Descriptors**
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well- supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

AO3